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A general pressure tensor calculation for molecular dynamics simulations

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A hybrid method for the calculation of the pressure tensor in molecular dynamics (MD) simulations, which combines both the thermodynamic and mechanical definitions, is presented here. The need for a new approach was motivated initially by MD simulations of crystalline poly(ethylene oxide) (PEO) (S. Neyertz, D. Brown, and J. O. Thomas, 1994, *J. chem. Phys.*, **101**, 10064), as the combination in this model of effectively 'infinite' chains, holonomic constraints and the long-range Coulomb potential calculated via the Ewald summation method, presented difficulties with regard to the traditional routes for calculation of the pressure tensor. However, this technique is quite general and thus applicable also to most MD simulations performed using periodic boundary conditions. In this paper, we address in detail the contribution of forces arising from different parts of the potential and the constraints to the pressure tensor. The method is then illustrated in a model of crystalline PEO, where the sensitivity of the pressure tensor to details of the potential provides an effective criterion for assessing possible parameter sets.

1. Introduction

Poly(ethylene oxide) (PEO), $-(\text{CH}_2\text{CH}_2\text{O})_n-$, has long been known in the field of polymer/salt electrolytes for its ability to dissolve a large variety of inorganic salts and exhibit high ionic conductivities [1]. Pure PEO and most PEO/salt systems generally show a mixed crystalline/amorphous character under ambient conditions, e.g. PEO is 70–85% crystalline at 300 K. Whereas ion mobility occurs mostly in the amorphous phase [2], a successful modelling of the PEO crystal structure was considered to be an essential first step in the development of a realistic potential for these systems. A molecular dynamics (MD) simulation of crystalline PEO was undertaken and reported recently [3].

In this study [3], results were presented for a model of PEO and a potential parameter set which accurately reproduces experimental and available X-ray refined diffraction data [4]. In selecting the optimized parameter set for these simulations, emphasis was placed on the behaviour of the pressure tensor $\mathbf{P}$. In principle, for a system at mechanical equilibrium, the off-diagonal elements of the pressure tensor should be zero, whereas the on-diagonal elements should all be equal and related to the ambient external hydrostatic pressure, $p = (P_{xx} + P_{yy} + P_{zz})/3$. However, in our model of crystalline PEO, there were three features which, in conjunction, caused problems with regard to the pressure tensor calculation. First, rigid holonomic constraints were used to remove high frequency vibrations from the system; second,

effectively 'infinite' PEO chains were created by bonding atoms in the primary chains to those in their periodic image; and third, electrostatic interactions were calculated using the Ewald summation method. The traditional route around the problems associated with systems including both rigid constraints and Ewald summations has been the so-called 'molecular' pressure tensor. However, this strategy requires a definition of the centre of mass (COM) at which to localize the momentum of a 'molecule'. The latter might be possible, but is by no means intuitive for infinite chains.

This paper addresses these problems in some detail and, by drawing from published sources, constructs a more straightforward and generally applicable method of calculating the pressure tensor. It is illustrated by reference to the crystalline PEO system for which it was originally developed. However, we emphasize that this is a general method, applicable to most classical MD simulations carried out in fully periodic systems.

In section 2 we give the main features of the model of crystalline PEO which served as a driving force for this new approach. A detailed description of the pressure tensor calculation is then presented in section 3. Examples of its usefulness in discriminating between potential models are given in section 4.

2. Details of the model

Extensive details of our model of PEO have been given previously [3], but for completeness we briefly outline the salient features.

X-Ray diffraction has shown that PEO crystallizes in helical chains with a period of two twists for every seven $-(\text{CH}_2\text{CH}_2\text{O})-$ monomer units [4]. The helicity arises as the OCCO angles adopt *gauche* conformations in the crystal, while the COCC angles are in the *trans* state. Equal numbers of fibres with TTG^- and TTG^+ sequences make up the structure. A schematic representation of the structure of crystalline PEO is shown in figure 1.

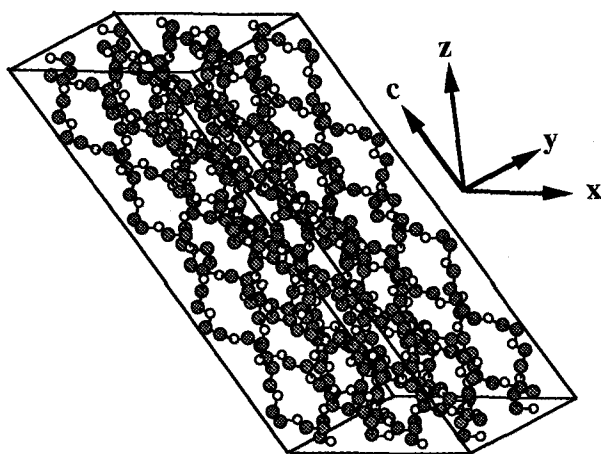


Figure 1. A schematic diagram of the initial configuration of crystalline PEO in the MD box. It is constructed from $2 \times 2 \times 2$ monoclinic crystallographic unit cells (cell parameters: $a = 8.05 \text{ \AA}$, $b = 13.04 \text{ \AA}$, c (fibre axis) $= 19.48 \text{ \AA}$, $\beta = 125.4^\circ$) and contains in total 1568 atoms in 16 chains although, for clarity, hydrogens are omitted here.

In the model of PEO described here, all bond lengths were rigidly constrained. Polymer backbone angle-bending deformations were described by a harmonic function in the bond angle θ ,

$$\Phi(\theta) = k_\theta/2 (\theta - \theta_0)^2, \quad (1)$$

and torsional motions around specific bonds by a sixth-order polynomial function in the cosine of the respective torsional angles τ ,

$$\Phi(\tau) = \sum_{j=0}^6 a_j (-1)^j \cos^j \tau. \quad (2)$$

The large number of degrees of freedom implied by the use of a full atomic model was reduced by applying the SHAKE routine described by Hammonds and Ryckaert to the hydrogen atoms [5, 6]; the scissoring, wagging and twisting C_nH_2 modes were constrained by holding the mid-point of the H–H vector on the bisector of the CC_nO bending angle, while the rocking modes were frozen by maintaining the H–H vector perpendicular to the vector joining the C and O atoms nearest to C_n .

Atoms belonging to the same polymer chain but separated by more than three bonds, as well as atoms belonging to different chains, were allowed to interact through the non-bonded potentials. Van der Waals interactions were described by the Buckingham potential,

$$\Phi_B(r_{ij}) = D \exp(-r_{ij}/\rho) - E/r_{ij}^6 \quad (3)$$

($r_{ij} = |r_{ij}|$), and Coulomb interactions by the electrostatic potential,

$$\Phi_C(r_{ij}) = \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} \quad (4)$$

where q_i and q_j are the atomic partial charges. The Ewald summation method [7] was used to evaluate the electrostatic interactions.

Periodic boundary conditions were applied in all three dimensions to avoid a large surface area/volume ratio. The influence of chain ends was reduced by creating effectively ‘infinite’ chains, in which the last carbon atom in a chain is bonded to the nearest periodic image of the first oxygen atom of the same chain.

3. Pressure calculation

Although there is no problem with the hydrostatic pressure, the system described in the previous section is of a type which renders itself particularly troublesome for the calculation of the individual elements of the pressure tensor. Difficulties are introduced through a combination of the geometric constraints, the use of Ewald summations to calculate the long-range Coulomb interactions, and chains being ‘tied’ across the periodic boundaries to make infinite molecules. To understand these difficulties, it is necessary first to recall the alternative strategies for the calculation of the pressure tensor.

3.1. Thermodynamic route

Problems introduced into molecular systems by a combination of rigid constraints and Ewald summations have been circumvented by calculating the so-called molecular

COM pressure tensor [8] within the framework of a thermodynamic definition of the pressure [9–11]. The scalar (hydrostatic) pressure can be defined thermodynamically as

$$p = -\left(\frac{\partial A}{\partial V}\right)_T = k_B T \left(\frac{\partial}{\partial V} \log Z_N(V, T)\right)_T, \quad (5)$$

where A is the Helmholtz free energy, $Z_N(V, T)$ is the canonical partition function, and V the volume. Given the COM positions $\{\mathbf{R}_i\}$, and the positions $\{\mathbf{r}_{i\alpha}\}$ of the atoms in a molecule, Smith [9, 10] has shown that, by re-writing the Hamiltonian of the fully periodic system in terms of scaled COM coordinates ($\mathbf{S}_i = \mathbf{R}_i/V^{1/3}$) and intramolecular distances ($\mathbf{d}_{i\alpha} = \mathbf{R}_i - \mathbf{r}_{i\alpha}$), both invariant under volume scaling, it is possible to carry out the necessary differentials which allow p to be calculated. The extension to a full calculation of the pressure tensor has also been given before [11, 12], and can be summarized in the following expression:

$$P_{\alpha\beta} = \frac{1}{V} \sum_{\gamma} \frac{\partial L}{\partial h_{\alpha\gamma}} h_{\beta\gamma}, \quad (6)$$

where L is the Lagrangian of the system, and $\mathbf{h} = (\mathbf{a}, \mathbf{b}, \mathbf{c})$ the matrix of basis vectors defining the shape and size of the MD box. Writing the Lagrangian in terms of $\{\mathbf{S}_i\}$ and $\{\mathbf{d}_{i\alpha}\}$ again avoids the problems of the rigid constraints and allows $\mathbf{P}$ to be calculated. However, as we believe there is no sensible definition of the COM for an effectively ‘infinite’ chain, this method is not directly applicable to the kind of system described in section 2.

3.2. Momentum flux

An alternative to the thermodynamic route outlined above is the mechanical definition of the pressure as a momentum flux across a virtual interface [13, 14]. By averaging over all possible interfaces of a periodic system, this definition leads to the familiar ‘virial’ expression [13]. A relevant application of this method to uncharged polymeric systems has been set out in detail previously [15]. In that approach, $\mathbf{P}$ was measured in an ‘atomic’ frame of reference, i.e. the momentum is localized at the positions of the atoms, and hence is symmetric; a more detailed discussion of the arbitrariness in the choice of momentum localization and its consequences can be found in [8]. For a system of N atoms in a fully periodic system of volume V , the pressure tensor can be evaluated from

$$\mathbf{P} = \frac{1}{V} \sum_{i=1}^N \left\{ \frac{1}{m_i} \mathbf{p}_i \mathbf{p}_i + \sum_{\substack{\text{all forces on } i \\ v}} \mathbf{r}_i^{(v)} \mathbf{f}_i^{(v)} \right\}, \quad (7)$$

where m_i and $\mathbf{p}_i$ are the mass and momentum of particle i , and the sum involving $\mathbf{r}_i^{(v)}$ and $\mathbf{f}_i^{(v)}$ implies a sum over all the individual force contributions $\mathbf{f}_i^{(v)}$ to the total force on i , where it must be emphasized that the position $\mathbf{r}_i^{(v)}$ is the same as that used in the calculation of that particular force. The notation $\mathbf{r}_i^{(v)}$ is a shorthand way of indicating the position used when calculating the v th contribution to the total force on i . Indeed, in simulations using periodic boundaries, the position of an atom when calculating the individual force contributions will not be invariant; it can be the primary position of particle i or of any of its periodic images. Equation (7) thus cannot be reduced to a single sum involving unique positions of the N atoms. It

should be noted also that the individual forces $f_i^{(v)}$ on atom i arise from all the potentials and the constraints in which atom i is involved. These forces may depend on the relative position of two particles (e.g. non-bonded pair forces and bonds), three particles (e.g. angle-bending), four particles (e.g. torsions) and even more (e.g. some constraints involving CH_2 groups). The detailed treatment of such terms is given in section 3.3 and will help to clarify the rather subtle nature of equation (7). Two additional points are worth noting. First, in the case where just two-body potentials are in operation, equation (7) reduces to the familiar double-sum over nearest-image separations and pair forces. Secondly, equation (7) is invariant with respect to the origin of coordinates. This follows trivially from the fact that the sum of all the forces is zero in the systems considered here.

In systems where Ewald summations are not used, the implementation of equation (7) in a MD simulation is straightforward [15]. In this particular case, there is no need to define a Lagrangian and thus the treatment of constraints poses no problems. However, in the system considered here and in many other cases, long-range Coulombic interactions are important, and difficulties will arise if these are calculated using the Ewald method.

In the Ewald method applied to our system, the total Coulombic energy is given by [16]

$$U_C = \frac{1}{2V\epsilon_0} \sum_{k \neq 0} A_k |Q_\Sigma|^2 + \frac{1}{4\pi\epsilon_0} \sum_{j=1}^N \sum_{m>j}^N \frac{q_j q_m}{|r_{jm}|} \text{erfc}(\alpha|r_{jm}|) - \frac{\alpha}{4\pi^{3/2}\epsilon_0} \sum_{j=1}^N q_j^2 - \frac{1}{4\pi\epsilon_0} \sum_{j=1}^N \sum_{m>j}^{m' \text{ bonded to } j} \frac{q_j q_m}{|r_{jm}|}, \quad (8)$$

where erfc is the complementary error function, α the Ewald separation parameter, r_{jm} the minimum image separation vector of the two atoms, the term A_k is given by

$$A_k = \frac{\exp(-k^2/4\alpha^2)}{k^2}, \quad (9)$$

and

$$Q_\Sigma = \sum_{j=1}^N q_j \exp(-ik \cdot r_j). \quad (10)$$

It should also be noted that

$$|Q_\Sigma|^2 = Q_\Sigma Q_\Sigma^* = \left(\sum_{j=1}^N q_j \exp(-ik \cdot r_j) \right) \left(\sum_{j=1}^N q_j \exp(ik \cdot r_j) \right), \quad (11)$$

i.e. Q_Σ^* is the complex conjugate of Q_Σ . The reciprocal space vectors k are defined with respect to the matrix of basis vectors h as

$$k = 2\pi(h^{-1})^t \begin{pmatrix} l \\ m \\ n \end{pmatrix}, \quad (12)$$

with l, m, n integers. In practice, the number of reciprocal space vectors included in the summation is limited by a judicious choice of α (see later), and by the use of an upper bound, K , i.e. $(l^2 + m^2 + n^2) \leq K^2$.

The double sum in the second term of equation (8) includes all possible pairs in the system. Together with the reciprocal space calculation (first term in equation (8)), it implies that all pairs have interacted through the full Coulomb potential. As in our model of PEO, atoms belonging to the same molecule are allowed to interact via the non-bonded part of the potential only if they are separated by more than three bonds; these interactions have to be subtracted from the total Coulombic contribution. They are removed by including the fourth term in equation (8).

The force on atom j due to the Coulombic potential is given by

$$\begin{aligned}
 f_j^c = \frac{-dU_C}{dr_j} = & \frac{-1}{2V\epsilon_0} \sum_{k \neq 0}^{\infty} A_k i k (Q_{\Sigma} q_j \exp(ik \cdot r_j) - Q_{\Sigma}^* q_j \exp(-ik \cdot r_j)) \\
 & + \frac{1}{4\pi\epsilon_0} \sum_{m \neq j}^N \frac{q_j q_m r_{jm}}{|r_{jm}|^2} \left(\frac{2\alpha}{\pi^{1/2}} \exp(-(\alpha|r_{jm}|)^2) + \frac{\text{erfc}(\alpha|r_{jm}|)}{|r_{jm}|} \right) \\
 & - \frac{1}{4\pi\epsilon_0} \sum_{\substack{m \text{ 'bonded' to } j \\ m \neq j}} \frac{q_j q_m}{4\pi\epsilon_0 |r_{jm}|^3} r_{jm}.
 \end{aligned} \tag{13}$$

Contributions to the total Coulombic force on a particle thus come from both real and reciprocal space sums. In reality, the reciprocal (or Fourier) space forces on a particular atom j are the remainder of the Coulombic force omitted from the real space sum, plus the forces which arise between j and all possible periodic images of all particles. Although the *total* Fourier space force on an atom can be evaluated, the individual pair contributions to this force cannot be separated. It is impossible to write an expression of the form required by equation (7).

3.3. Hybrid pressure calculation

For the above reasons, a hybrid method for the pressure tensor calculation has been adopted. In this new approach, the mechanical definition (equation (7)) is used, but the thermodynamic definition (equation (6)) is resorted to for those forces (i.e. the Fourier space part of the Ewald sum) where the former is not applicable. An alternative, and equally justifiable, viewpoint is that equation (6) is used in an atomic frame of reference except for those forces not derivable from a continuous potential (i.e. the constraint forces) for which we use equation (7). It should be noted that, in principle, one could use either the thermodynamic or mechanical definitions alone. However, for equation (7), this would mean replacing the Ewald sum by a straight pair summation; this is possible though not recommended on account of the increased computational cost. For equation (6), the rigid constraints would have to be incorporated into the Lagrangian, which is by no means obvious, although some progress in this area has been made [17]. The hybrid method presented below is thus a compromise between the two definitions of the pressure. In what follows, the contribution of forces arising from different parts of the potential and the constraints to the pressure tensor are discussed in turn.

3.3.1. Constraint forces

Following the evaluation of all the forces from the continuous potentials at a particular time t in an MD simulation, it is customary [15] to integrate the equations of motion forward using a finite difference algorithm to a time $t + \Delta t$ in the absence of the yet undetermined constraint forces in order to obtain the 'free flight' positions

$r_i^0(t + \Delta t)$. The various coupled constraint equations are then solved, typically using a SHAKE [6] type iterative procedure, to determine the actual new positions $r_i(t + \Delta t)$. The number $n(t)$ of such iterations required at a particular time step depends on the system conditions, the complexity of the constraints applied, and the required tolerance. The difference between the final positions and the free-flight positions can then be related to the total force of constraint on an atom

$$g_i(t) = \frac{m_i \{r_i(t + \Delta t) - r_i^0(t + \Delta t)\}}{\Delta t^2}. \quad (14)$$

If during the SHAKE procedure, only positions of atoms which preserve the connectivity of the molecule [15] were used, i.e. no periodic image transformations required, the contribution of the constraint forces to the pressure tensor from equation (7) can be written simply as

$$P_g(t) = \frac{1}{V} \sum_{i=1}^N r_i(t) g_i(t). \quad (15)$$

In the more complicated situation where molecules are rendered effectively infinite in extent by rigid bonds between atoms in the primary chain and images of atoms in the primary chain, as described in our model of crystalline PEO, much care has to be taken in using the relevant position (primary or image) of an atom when applying equation (7).

Consider the individual application of each of the N_c constraints in the system in turn. The number n_c of atoms involved in a particular constraint varies according to its type. For a bond length constraint $n_c = 2$, but more complicated constraints, like those restricting hydrogen modes [6], can directly affect up to five atoms. When considering the l th constraint, the positions of the n_c^l atoms involved must form a connected set $\{r_i(t)\}^l$ (e.g. for a bond length constraint $|r_i(t) - r_j(t)|$ must equal the requisite bond length). It follows that i' will be either the primary position of i or that of one of its images, depending on the relative position of i' and the other atoms in the set. At the j th application of this l th constraint in the iterative procedure, following the free-flight phase, a current set of unconstrained positions $\{r_i^{j-1}(t + \Delta t)\}^l$ at time $t + \Delta t$ (i.e., those following all previous $j - 1$ applications of all N_c constraints, and the $l - 1$ constraints already applied at the current (j th) iteration) are SHAKEn to give a new set of constrained positions $\{r_i^j(t + \Delta t)\}^l$. From these two sets of positions it is possible to determine (cf. equation (14)) the contribution $\Delta g_i^j(t)$ of the j th application of the l th constraint to the total force of constraint using the expression

$$\Delta g_i^j(t) = \frac{m_i \{r_i^j(t + \Delta t) - r_i^{j-1}(t + \Delta t)\}}{\Delta t^2}, \quad (16)$$

where it should be noted that the prime is dropped on the i in $\Delta g_i^j(t)$ as the force applies to i and all its images. The sum of all such force contributions over all the $n(t)$ iterations and all N_c constraints at a particular time step gives, of course, $\{g_i(t)\}$ as in equation (14). However, as the coordinates used in evaluating that contribution to the force are known individually, equation (7) can now be implemented:

$$P_g(t) = \frac{1}{V} \sum_{j=1}^{n(t)} \sum_{l=1}^{N_c} \sum_i^{\text{all } n_c^l \text{ atoms}} r_i(t) \Delta g_i^j(t). \quad (17)$$

Evaluation of equation (17) during a simulation presents no problems as it stands,

and trials on systems containing molecules of finite extent proved that it gave the same results as equation (15) to as high an order as could be expected from the different round-off errors. It was also found to give correct results when molecules from such a system were artificially 'broken' by using as primary coordinates those initially within the MD box rather than the original ('unfolded') ones which preserved the connectivity. Satisfactory checks were made as well to ensure that the calculation was invariant to the completely arbitrary positions of the periodic boundaries. Finally, it should be noted that computational savings were made in the simulations of crystalline PEO reported here by using 'ghost' particles, i.e. images of particles involved in constraints straddling periodic boundaries. This allowed the $\Delta g_i^j(t)$ to be associated with a particular invariant atom or ghost atom and thus the pressure tensor to be calculated using an expression akin to equation (15), but with the sum now extending over the total number of atoms plus ghost atoms.

3.3.2. Bending forces

For the bending forces, the contribution to the pressure tensor is computed as

$$\mathbf{P}_\theta = \frac{1}{V} \sum_{\text{All } \theta} \mathbf{r}_i \mathbf{f}_i^\theta + \mathbf{r}_j \mathbf{f}_j^\theta + \mathbf{r}_k \mathbf{f}_k^\theta, \quad (18)$$

where atoms i, j and k define a particular bond angle θ , and the positions of these three atoms *must* be those used in calculating the corresponding forces $\mathbf{f}^\theta$, i.e. the primary coordinates of a particle or any of its periodic images. It should be noted that, as the sum of the forces and the sum of the torques for this form of multi-particle potential are both zero, equation (18) is invariant to the origin of coordinates and the resultant pressure tensor is thus symmetric. Another point to note is that in practice it is not always correct to assume that the contribution of the bending forces to the trace of the pressure tensor is zero [18]. This is only true if one derives the equations for the bending forces on the particles assuming that the bonds are flexible. Making use of the rigidity of the bonds to simplify the algebra† also means that the bending contribution has to be included. In any case, for a general calculation of $\mathbf{P}$, all forces must be taken into account.

3.3.3. Torsional forces

For the torsional forces, the contribution to the pressure tensor is computed in a similar way as for the bending forces, i.e.

$$\mathbf{P}_\tau = \frac{1}{V} \sum_{\text{All } \tau} \mathbf{r}_i \mathbf{f}_i^\tau + \mathbf{r}_j \mathbf{f}_j^\tau + \mathbf{r}_k \mathbf{f}_k^\tau + \mathbf{r}_l \mathbf{f}_l^\tau, \quad (19)$$

where atoms i, j, k and l define a particular torsion angle τ , and the positions of these four atoms *must* be those used in calculating the corresponding forces $\mathbf{f}^\tau$. Similar remarks to those for bends also apply.

3.3.4. Buckingham forces

As this pair potential is relatively short in range, its contribution can be calculated in the usual way,

$$\mathbf{P}_B = \frac{1}{V} \sum_{i=1}^N \sum_{\substack{j=1 \\ j \text{ not 'bonded' to } i}}^N \mathbf{r}_{ij} \mathbf{f}_{ij}^B, \quad (20)$$

† Following a request by Peter Ahlström a note will be submitted to *Molecular Simulation* regarding this simplification of the algebra.

where r_{ij} is the minimum image separation vector, and f_{ij}^B the force from the Buckingham potential on particle i due to particle j . The double sum extends over only those pairs deemed to interact via the Buckingham potential. To account for the truncation of the potential at a finite range, a long-range correction to the pressure is calculated on the basis that $g(r) = 1$ beyond the cutoff [7]. This correction is assumed to be isotropic and is added to each of the on-diagonal components. Long-range corrections to the off-diagonal elements disappear through symmetry.

3.3.5. Coulombic forces

Before making a complete calculation of the individual elements of the Coulombic contribution to the pressure tensor, its contribution to the trace of the pressure tensor alone is first considered. This illustrates a well known point and will serve later as a useful check of the full Coulombic calculation. The Coulombic contribution to the hydrostatic pressure is defined as

$$p_C = \frac{P_{C_{xx}} + P_{C_{yy}} + P_{C_{zz}}}{3} = \frac{1}{3V} \sum_{i=1}^N \sum_{\substack{j>i \\ j \text{ not 'bonded' to } i}}^{\infty} r_{ij} \cdot f_{ij}^C \quad (21)$$

where, in the sum over j , only those atoms with which i interacts through the Coulomb potential are included. For example, in the model of PEO considered in this paper, the 'bonded' interactions between pairs separated by three or fewer bonds are excluded. The $1/r$ nature of the Coulombic energy means that the number of j to be taken is effectively infinite. However, the Coulombic force on i owing to j is given by

$$f_{ij}^C = \frac{-d\Phi_C(|r_{ij}|)}{dr_i} = \frac{q_i q_j}{4\pi\epsilon_0 |r_{ij}|^3} r_{ij}, \quad (22)$$

so substituting for f_{ij}^C in equation (21) returns the familiar result

$$p_C = \frac{1}{3V} \sum_{i=1}^N \sum_{\substack{j>i \\ j \text{ not 'bonded' to } i}}^{\infty} \frac{q_i q_j}{4\pi\epsilon_0 |r_{ij}|} = \frac{1}{3V} U_C, \quad (23)$$

where U_C is imply the total Coulombic energy. As U_C is calculated via the Ewald summation method (equation (8)), the above expression provides a consistency check (see section 4.3) on the more complete calculation of the individual elements of the pressure tensor to be presented now.

The contribution of the second and fourth terms in equation (8) to the pressure tensor can be calculated directly from the forces which arise and the separation vectors. Denoting the sum of these two 'real space' terms as U_C^R , we obtain their contribution to the pressure tensor as

$$P_C^R = \frac{1}{V} \sum_{i=1}^N \sum_{j>i}^N r_{ij} \left\{ \frac{-dU_C^R}{dr_{ij}} \right\}, \quad (24)$$

where r_{ij} is again the minimum image vector, and the Ewald separation parameter α has been chosen such that U_C^R converges within a distance less than or equal to half the smallest perpendicular distance between any two of the MD cell faces. The third term in equation (8), often referred to as the 'self-correction' term, does not depend on position and so does not produce forces. However, the reciprocal, or Fourier, space term (first term in equation (8)) does depend on position and produces

forces on particles. As was explained in section 3.2, although there is an expression for the *total* Fourier space force on an atom, it is impossible to resolve it into individual atom-atom contributions and, hence, to write an expression of the form of equation (24). For this reason the thermodynamic definition of the pressure tensor, equation (6), is used to obtain the Fourier space force contribution.

Applying equation (6) to the relevant part of the Lagrangian gives the required contribution as

$$P_{\alpha\beta}^F = \frac{1}{V} \sum_{\gamma} \frac{-\partial U_C^F}{\partial h_{\alpha\gamma}} h_{\beta\gamma}, \quad (25)$$

where U_C^F is defined as

$$U_C^F = \frac{1}{2V\epsilon_0} \sum_{\mathbf{k} \neq 0} A_{\mathbf{k}} |Q_{\Sigma}|^2, \quad (26)$$

i.e. the Fourier space contribution to the total Coulombic energy. The full derivation is given in the appendix and the final result is the following expression:

$$P_{\alpha\beta}^F = \frac{1}{2V^2\epsilon_0} \sum_{\mathbf{k} \neq 0} A_{\mathbf{k}} |Q_{\Sigma}|^2 \left\{ \delta_{\alpha\beta} - \left(\frac{1}{4\alpha^2} + \frac{1}{k^2} \right) 2k_{\alpha} k_{\beta} \right\}, \quad (27)$$

where $\delta_{\alpha\beta}$ is the usual Kronecker delta.

3.3.6. Full expression for the pressure tensor

Finally, combining all the various contributions discussed above with the kinetic component gives the full expression for the pressure tensor

$$\mathbf{P} = \left\{ \frac{1}{V} \sum_{i=1}^N \frac{1}{m_i} \mathbf{p}_i \mathbf{p}_i \right\} + \mathbf{P}_g + \mathbf{P}_{\theta} + \mathbf{P}_t + \mathbf{P}_B + \mathbf{P}_C^R + \mathbf{P}_C^F. \quad (28)$$

Although the expression given here is for a particular force field, the treatment of other forms of potential follows straightforwardly. It is important to stress that this approach is quite general in nature.

4. Case study: potential screening in poly(ethylene oxide)

As mentioned earlier, this new approach to the pressure tensor calculation was motivated by a constant volume molecular dynamics simulation of crystalline PEO [3]. We now give examples of how different parameter sets can be screened by applying the pressure tensor calculation described above. Illustrated here are different potential sets for the torsional, Buckingham and Coulombic interactions in PEO.

4.1. Invariant details of the simulations

Simulations were carried out at constant volume for a system of $2 \times 2 \times 2$ monoclinic crystallographic unit cells (cell parameters: $a = 8.05 \text{ \AA}$, $b = 13.04 \text{ \AA}$, c (fibre axis) $= 19.48 \text{ \AA}$, $\beta = 125.4^\circ$). The MD box thus contained 1568 atoms in 16 chains in a volume of $13\,335.5 \text{ \AA}^3$, i.e. a density of 1.229 g cm^{-3} .

Bond lengths were constrained rigidly to $1.43 \text{ \AA}$ for C–O, $1.54 \text{ \AA}$ for C–C, and $1.09 \text{ \AA}$ for C–H. All degrees of freedom involving hydrogens were frozen out, as stated

earlier, with HCH angles fixed at 109.5° . The flexible backbone valence angles had equilibrium bond angles θ_0 of 110° for OCC, and 112° for COC. The above geometrical information is consistent with that used in the original refinement procedure [4]. The backbone bending force constants k_θ used in equation (1) were taken from [19] and are $152.9 \text{ kcal mol}^{-1} \text{ rad}^{-2}$ and $219.1 \text{ kcal mol}^{-1} \text{ rad}^{-2}$ for the OCC and COC angles, respectively.

The Buckingham potential was truncated at $6.56 \text{ \AA}$, which corresponds to half the shortest perpendicular distance between two faces of the simulation cell. Long-range corrections to the energy and the pressure were calculated on the basis of $g(r) = 1$ beyond the cutoff. Optimal convergence of the Ewald sum within acceptable CPU costs was obtained using $\alpha = 0.4 \text{ \AA}^{-1}$ and $K = 13$. The real space potential was also truncated at $6.56 \text{ \AA}$.

The leap-frog algorithm was used to integrate the equations of motion with a timestep $\Delta t = 10^{-15} \text{ s}$. Systems were kept at 300 K by weak coupling to a heat bath [20] with a relaxation time $\tau_T = 1 \text{ ps}$. Verlet-type neighbour lists were created at the beginning of each run with the truncation value of the non-bonded potentials taken as the radius of the inner sphere, and a shell of width of $1.5 \text{ \AA}$. These lists were updated automatically when any atom had diffused a distance greater than or equal to half the shell width [15]. Lists of atoms interacting through bonded potentials were also created initially and used throughout the simulations. The relative tolerance for geometrical constraints was set to 10^{-6} [6] which led to ~ 17 SHAKE iterations per step. Tests of the basic integration algorithm have already been given [3].

All simulations presented here were performed for 20 ps with the initial configuration being taken from the end of the previous 120 ps simulation of crystalline PEO [3]. In each case, the first 5 ps were discarded from the averages.

4.2. Parameter sets in potential trials

Ab initio calculations at the MP2/6-311++G**//HF/3-21G level performed on diglyme $\text{CH}_3\text{O}(\text{CH}_2\text{CH}_2\text{O})_2\text{CH}_3$ [21] were used as a basis for parametrizing the torsional potentials. The set of coefficients a_j for $j = 0-6$ obtained from least-squares fits of the *ab initio* data to equation (2) for both CC and CO dihedral angles τ -A is given in table 1. For comparison, alternative sets fitted to the same data with fifth and fourth order polynomials were made. These sets, referred to respectively as τ -B and τ -C, are also shown in table 1.

Two sets of the Buckingham parameters (D , ρ and E) are displayed in table 2.

Table 1. Coefficients a_j used in equation (2) for the different torsional potential sets. All values are given in kcal mol^{-1} .

Set	Centre bond	a_0	a_1	a_2	a_3	a_4	a_5	a_6
τ -A	C-C	0.528	3.629	4.262	-7.753	-3.313	-0.284	2.943
	C-O	1.238	1.340	1.498	-3.685	-0.162	-1.091	0.852
τ -B	C-C	0.621	3.629	2.601	-7.753	1.110	-0.284	0
	C-O	1.265	1.340	1.017	-3.685	1.119	-1.091	0
τ -C	C-C	0.621	3.719	2.601	-8.109	1.110	0	0
	C-O	1.265	1.683	1.017	-5.051	1.119	0	0

Table 2. The two sets of parameters for the Buckingham form of the van der Waals potential (equation (3)): D is given in kcal mol⁻¹, ρ in Å, and E in kcal mol⁻¹ Å⁶.

Atom pairs	vdW-A			vdW-B		
	D	ρ	E	D	ρ	E
O···O	58 298.9	0.2485	192.1	223 000.0	0.2179	245.0
O···C	42 931.6	0.2755	352.8	438 500.0	0.2179	294.0
O···H	20 432.6	0.2445	98.8	41 850.0	0.2188	96.5
C···C	31 615.1	0.3025	647.8	908 600.0	0.2179	363.0
C···H	15 046.7	0.2715	181.5	86 100.0	0.2188	127.0
H···H	7 161.2	0.2405	50.8	9 950.0	0.2203	45.2

Table 3. Partial charges q_i used in the electrostatic potential (equation (4)).

Atom type	Partial charges/ e		
	q -A	q -B	q -C
O	-0.3480	-0.2680	-0.2800
C	0.1030	0.0440	-0.0200
H	0.0355	0.0450	0.0800

The first, vdW-A, was derived from the universal force field (UFF) [22]. The second, vdW-B, is an older one due to Abe and Mark [23].

Three different sets of atomic partial charges q_i were also tested. The first, q -A, were taken from the *ab initio* study of 1,2-dimethoxyethane [24]. Alternative sets q -B and q -C were determined from semi-empirical single-point MNDO and AM1 calculations, respectively, performed on the 7-monomer asymmetric unit of crystalline PEO. All three sets of charges are given in table 3.

Comparative simulations were based on various combinations of the above sets of parameters. Details of those permutations selected are presented in table 4.

4.3. Pressure calculation consistency check

Before discussing the results of the potential trials described above, we first illustrate the internal consistency check alluded to in section 3.3.5. Using the initial PEO crystalline configuration and parameter set combination 1 (table 4), short calculations were made to study the convergence of the Ewald summation with respect to the parameter K , i.e. that controlling the number of $\mathbf{k}$ vectors included in the Fourier space sum. In table 5, the values of the Fourier space contributions to the hydrostatic pressure p_C^F , and the total Coulombic energy U_C^F are given as functions of increasing K . p_C^F can be calculated in two ways which constitute a check on the proper convergence of the Ewald summation. Equation (27) gives a direct route to this contribution

$$p_{C\text{direct}}^F = \frac{1}{3} \text{Tr} \{ \mathbf{P}_C^F \}, \tag{29}$$

but an indirect calculation is also available by using equations (23) and (24),

Table 4. Average results for the scalar pressure p , pressure tensor components $P_{\alpha\beta}$ and energy for various combinations of the potential sets described in section 4.2. The rows labelled τ , vdW and q indicate, respectively, which sets of torsion parameters, van der Waals (Buckingham) parameters and partial charges were used. Pressures are given in MPa. The bond angle (U_θ), torsion angle (U_t), van der Waals (U_B), Coulombic (U_C) and total potential (U_ϕ) energies are given in kcal mol⁻¹ (per system).

	Parameter set combinations					
	1	2	3	4	5	6
τ	A	B	C	A	A	A
vdW	A	A	A	B	A	A
q	A	A	A	A	B	C
p	-4 ± 8	-29 ± 9	-35 ± 11	-791 ± 14	128 ± 11	54 ± 12
P_{xx}	12 ± 4	-9 ± 6	-8 ± 5	-784 ± 10	145 ± 5	80 ± 7
P_{yy}	31 ± 6	30 ± 6	29 ± 9	-754 ± 8	124 ± 8	23 ± 8
P_{zz}	-56 ± 3	-108 ± 5	-125 ± 5	-833 ± 7	115 ± 6	58 ± 6
P_{xy}	0 ± 3	-1 ± 3	-1 ± 3	0 ± 5	-2 ± 2	1 ± 3
P_{xz}	5 ± 2	27 ± 2	32 ± 4	-6 ± 3	-71 ± 3	-83 ± 3
P_{yz}	-2 ± 2	0 ± 3	-1 ± 2	3 ± 4	2 ± 2	-2 ± 3
U_θ	187.1 ± 0.8	183.4 ± 1.2	184.7 ± 1.0	186.2 ± 0.8	182.7 ± 1.3	186.6 ± 0.8
U_t	104.9 ± 0.5	93.6 ± 0.4	112.6 ± 0.5	119.2 ± 0.5	107.2 ± 0.5	105.8 ± 0.6
U_B	-1439.2 ± 0.7	-1439.7 ± 0.9	-1437.9 ± 1.0	-1455.5 ± 0.9	-1437.0 ± 1.0	-1442.1 ± 0.9
U_C	-1147.1 ± 0.1	-1148.8 ± 0.2	-1149.2 ± 0.2	-1148.4 ± 0.3	-408.0 ± 0.1	-3.0 ± 0.1
U_ϕ	-2294.3 ± 0.4	-2311.6 ± 0.3	-2289.9 ± 0.4	-2298.4 ± 0.5	-1555.2 ± 0.3	-1152.6 ± 0.3

Table 5. The convergence of the Fourier space contributions to the hydrostatic pressure p_C^F and the Coulombic energy U_C^F . The parameter K limits the number of k vectors included in the summation (see text for details). The subscripts direct and indirect refer to the two calculations discussed in section 4.3. The pressures are given in MPa and U_C^F is given in kcal mol⁻¹ (per system).

K	$p_{C_{\text{direct}}}^F$	$p_{C_{\text{indirect}}}^F$	U_C^F
0	0	-101.52	0
2	0.22	-101.08	2.52
4	-9.00	-97.91	20.75
6	-72.58	-84.70	96.81
8	-79.79	-83.70	102.60
10	-81.78	-83.50	103.75
11	-82.62	-83.42	104.21
12	-82.83	-83.40	104.30
13	-83.14	-83.38	104.44
14	-83.17	-83.38	104.45

$$p_{C_{\text{indirect}}}^F = \frac{1}{3V} U_C - \frac{1}{3} \text{Tr} \{ \mathbf{P}_C^R \}. \quad (30)$$

The results given in table 5 demonstrate the rapid convergence of these two calculations, thus confirming simultaneously the internal consistency of both the real and reciprocal space parts of the Ewald summation. Not surprisingly, the rate of convergence coincides with that of U_C^F . The use of $K = 13$ in the MD simulations reported here is justified, given the relatively small difference between the two calculations of p_C^F compared with the typical statistical errors expected in the pressure (see table 4).

4.4. Results and discussion

It is clear from table 4 that some components of the pressure tensor are very sensitive to changes in potential parameters. A most striking illustration comes from a comparison of combinations 1, 2 and 3, that differ only by the order of the polynomial fitted to the experimental data for the torsional potential [21] (respectively sixth, fifth and fourth order). Whereas it is well known that the higher order of the fitted polynomial, the better the precision of the fit, differences between τ -A, τ -B and τ -C potential curves are not really noticeable if the entire 360° range for the torsional angle is plotted (figures not shown). However, from our previous study of crystalline PEO (see figure 5 of [3]), it is known that only the regions within $\pm 30^\circ$ of the minima in the *gauche* wells are populated for OCCO angles and similarly for the *trans* well for CCOC angles. Therefore, in figures 2 and 3 these significant portions of the potential are plotted. It can be seen that the curves fitted to the rotation energy around the CC bonds are shifted with respect to each other, and thus have minima at slightly different angles ($\sim 75^\circ$ for τ -A, and $\sim 73^\circ$ for both τ -B and τ -C). This is an important feature in the model of PEO as this angle effectively controls the 'tightness' of the helical coil. Ordinarily, changes in the position of the minimum would give rise to shorter or longer equilibrium lengths of the chains, but in our model of crystalline PEO, where the lengths and the numbers of turns of the helices

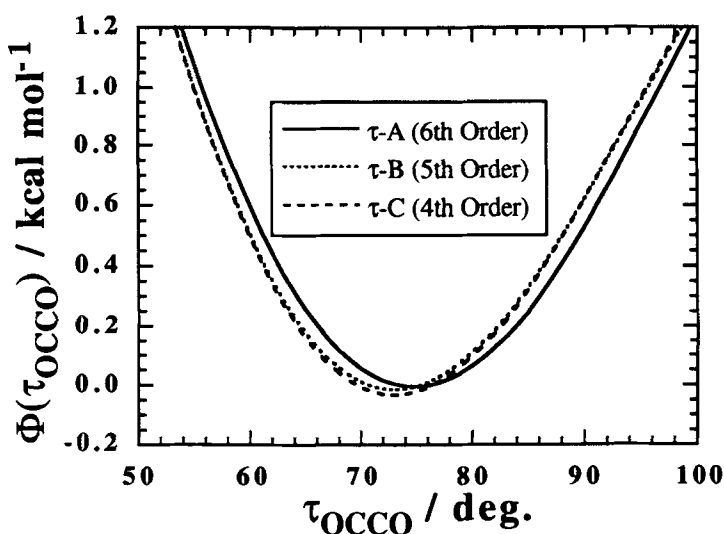


Figure 2. The different torsional potentials used to describe restricted rotation of the OCCO angles (see table 1). Only the variation in the G^- well is shown as in the PEO crystal all OCCO angles exist in this, or the symmetric, G^+ state. Note the shift in the position of the minimum with the order of the polynomial fit.

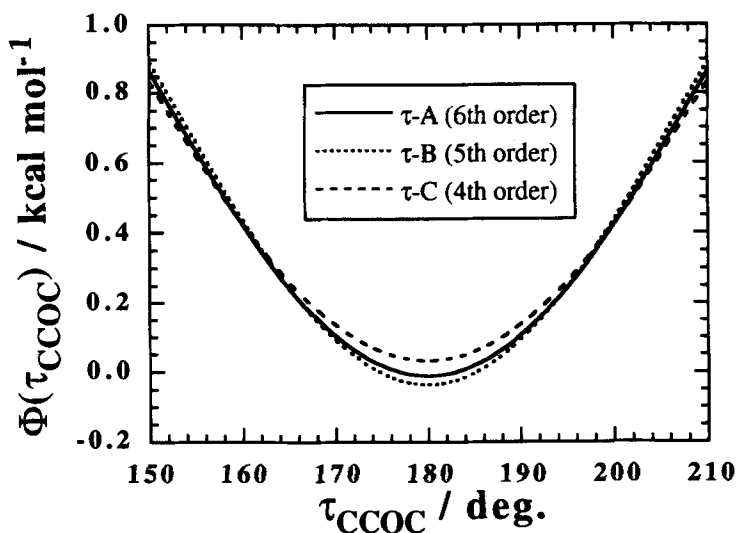


Figure 3. As figure 2 for CCOC angles. Only the *trans* well is shown as all CCOC angles remain in this well throughout the simulation. The potentials as defined have to be symmetric about 180° .

are imposed artificially through the combined use of periodic boundary conditions and effectively infinite chains, this manifests itself as a change in the tension in the coils. For the geometry of the system, the chain axes are orientated at 35.4° to the z axis in the xz plane (see figure 1); this translates into a clearly observable influence on P_{zz} and P_{xx} which show changes consistent with an increase in tension in the helical coils on going from τ -A to τ -B and τ -C. Of the other components, only P_{xx}

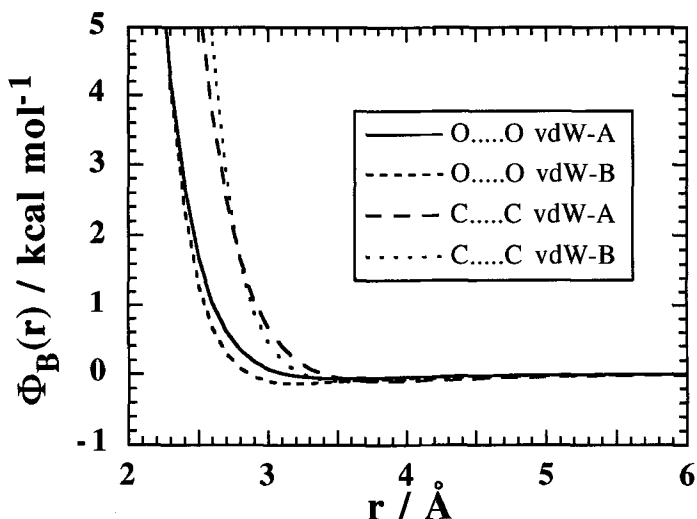


Figure 4. The two sets of Buckingham potentials for $\text{O} \cdots \text{O}$ and $\text{C} \cdots \text{C}$ non-bonded van der Waals interactions. The key to the parameter sets used in generating these curves from equation (3) is as given in table 2.

shows significant changes and these too are consistent with an increase in the coil tension. As cell size and shape are usually determined with precision in X-ray diffraction studies, the behaviour of the pressure tensor confirms the better suitability of combination 1 over 2 or 3 for our model of crystalline PEO.

The comparison of combinations 1 and 4 exemplifies the influence of the Buckingham parameters on the pressure tensor components. The change in the total van der Waals energy (U_B), shown in table 4, appears relatively small, and indeed the potential curves shown in figures 4 and 5, of vdW-A and vdW-B, are similar. The huge values of the on-diagonal elements, however, indicate clearly that the

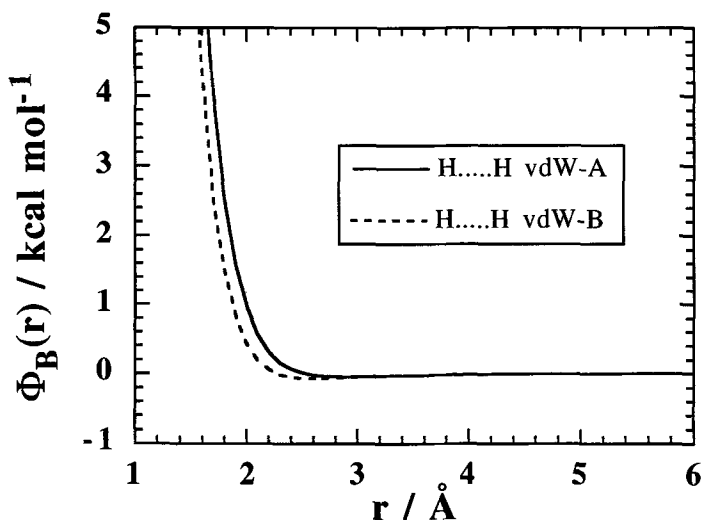


Figure 5. As figure 4 for $\text{H} \cdots \text{H}$ non-bonded van der Waals interactions.

simulation box in combination 4 very much wants to contract and that vdW-B is inappropriate for our simulations.

Also displayed in table 4 are results for parameter sets in which only the partial charges are changed with respect to each other, i.e. combinations 1, 5 and 6. There is a general loss in Coulombic cohesive energy U_C for combinations 5 and 6, which takes place with only small changes in the other energies. Concomitant with this loss of cohesion is an increase in isotropic pressure. A comparison of q -A and q -B suggests that this change results primarily from the charges on the backbone atoms being reduced. More detail is obtained by looking at the P_{xz} component, which shows in combinations 5 and 6 a tendency for the fibres to want to extend themselves. The changes on going from q -B to q -C are less obvious to interpret but give a good indication of the sensitivity of the system to changes in the charge distribution. In any case, it appears clear that the set of partial charges q -A, which have been obtained by a quantum mechanical method [24], should be chosen over q -B and q -C, which have been calculated by more crude MNDO and AM1 semi-empirical single-point methods.

5. Conclusion

By combining the mechanical and thermodynamic routes to the pressure tensor, we have derived a hybrid calculation which is generally applicable to fully periodic systems. The power of the method has been demonstrated by applying it to a system, crystalline PEO, that presented difficulties for traditional routes of calculating the pressure tensor. As a consequence of this, it has been possible to distinguish the effect of different parts of the potential on the pressure tensor and to identify, for example, those contributions that have the most influence. Most interesting is the fact that pressure tensor components as such are already far more sensitive probes to a small change in interaction potential than energies. While potential space-searching is still limited by computational resources and the situation complicated by the huge number of possible combinations of parameters, the PEO case study illustrates the sensitivity and usefulness of the calculation of the pressure tensor components in assessing possible parameters sets.

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Appendix

Starting from equations (25) and (26) given in the main text, a detailed derivation of equation (27), the contribution to the pressure tensor of the Fourier space part of the Ewald sum, is given here. From equation (25) we have

$$P_{\alpha\beta}^F = \frac{1}{V} \sum_{\gamma} \frac{-\partial U_C^F}{\partial h_{\alpha\gamma}} h_{\beta\gamma} = \frac{-\partial U_C^F}{\partial h_{\alpha x}} h_{\beta x} + \frac{-\partial U_C^F}{\partial h_{\alpha y}} h_{\beta y} + \frac{-\partial U_C^F}{\partial h_{\alpha z}} h_{\beta z}. \quad (\text{A } 1)$$

Using the definition of U_C^F given in equation (26), the partial differentials in equation (A 1) can be expanded to

$$\frac{-\partial U_C^F}{\partial h_{\alpha\gamma}} = \frac{-1}{2\epsilon_0} \left\{ \frac{\partial V^{-1}}{\partial h_{\alpha\gamma}} \sum_{\mathbf{k} \neq \mathbf{0}} A_k |Q_\Sigma|^2 + \frac{1}{V} \sum_{\mathbf{k} \neq \mathbf{0}} \left[\frac{\partial A_k}{\partial h_{\alpha\gamma}} |Q_\Sigma|^2 + A_k \left(\frac{\partial Q_\Sigma}{\partial h_{\alpha\gamma}} Q_\Sigma^* + Q_\Sigma \frac{\partial Q_\Sigma^*}{\partial h_{\alpha\gamma}} \right) \right] \right\}. \quad (\text{A } 2)$$

Using the relation $(\partial V / \partial h_{\alpha\gamma}) = V h_{\gamma\alpha}^{-1}$ [11] and gathering together terms of the first kind in equation (A 2) gives

$$\frac{1}{2\epsilon_0 V^2} \sum_{\mathbf{k} \neq \mathbf{0}} A_k |Q_\Sigma|^2 \sum_\gamma h_{\gamma\alpha}^{-1} h_{\beta\gamma} = \frac{1}{2\epsilon_0 V^2} \sum_{\mathbf{k} \neq \mathbf{0}} A_k |Q_\Sigma|^2 \delta_{\alpha\beta}. \quad (\text{A } 3)$$

The latter equality follows from the definition of the inverse, $\mathbf{h}\mathbf{h}^{-1} = \mathbf{1}$.

If one considers the second term in equation (A 2), it follows from equation (9) that

$$\frac{\partial A_k}{\partial h_{\alpha\gamma}} = -2\mathbf{k} A_k \left(\frac{1}{4\alpha^2} + \frac{1}{k^2} \right) \frac{\partial \mathbf{k}}{\partial h_{\alpha\gamma}}. \quad (\text{A } 4)$$

To differentiate the term in $\mathbf{k}$, we recall its definition given in equation (12) which we rewrite as

$$\mathbf{k} = 2\pi(\mathbf{h}^{-1})^t \mathbf{K} \quad (\text{A } 5)$$

with $\mathbf{K} = (l, m, n)$. Components of $\mathbf{k}$ can then be written as

$$k_\nu = 2\pi \sum_\mu h_{\mu\nu}^{-1} K_\mu. \quad (\text{A } 6)$$

Now, using the relationship [11]

$$\frac{\partial h_{\mu\nu}^{-1}}{\partial h_{\alpha\gamma}} = -h_{\mu\alpha}^{-1} h_{\gamma\nu}^{-1}, \quad (\text{A } 7)$$

the differential of equation (A 6) with respect to $h_{\alpha\gamma}$ becomes

$$\frac{\partial k_\nu}{\partial h_{\alpha\gamma}} = 2\pi \sum_\mu \frac{\partial h_{\mu\nu}^{-1}}{\partial h_{\alpha\gamma}} K_\mu = -2\pi h_{\gamma\nu}^{-1} \sum_\mu h_{\mu\alpha}^{-1} K_\mu = -h_{\gamma\nu}^{-1} k_\alpha, \quad (\text{A } 8)$$

with the last equality arising from equation (A 6). It thus follows that

$$\begin{aligned} \sum_\gamma \frac{\partial A_k}{\partial h_{\alpha\gamma}} h_{\beta\gamma} &= 2A_k \left(\frac{1}{4\alpha^2} + \frac{1}{k^2} \right) k_\alpha \mathbf{k} \cdot \left(\sum_\gamma h_{\gamma x}^{-1} h_{\beta\gamma}, \sum_\gamma h_{\gamma y}^{-1} h_{\beta\gamma}, \sum_\gamma h_{\gamma z}^{-1} h_{\beta\gamma} \right) \\ &= 2A_k \left(\frac{1}{4\alpha^2} + \frac{1}{k^2} \right) k_\alpha \mathbf{k} \cdot \hat{\beta} = 2A_k \left(\frac{1}{4\alpha^2} + \frac{1}{k^2} \right) k_\alpha k_\beta, \end{aligned} \quad (\text{A } 9)$$

where $\hat{\beta}$ is a unit vector in the β direction and the relation $\mathbf{h}\mathbf{h}^{-1} = \mathbf{1}$ has again been used.

Finally, the last terms in equation (A 2) involve differentials of Q_Σ and Q_Σ^* with respect to elements of $\mathbf{h}$. For example, using the definition of Q_Σ (equation (10)) gives

$$\frac{\partial Q_\Sigma}{\partial h_{\alpha\gamma}} = \sum_{j=1}^N q_j (-i) \exp(-i\mathbf{k} \cdot \mathbf{r}_j) \frac{\partial(\mathbf{k} \cdot \mathbf{r}_j)}{\partial h_{\alpha\gamma}}. \quad (\text{A } 10)$$

However, it is relatively straightforward to show that the dot product $\mathbf{k} \cdot \mathbf{r}_j$ is invariant to changes in $\mathbf{h}$, so these latter terms drop out. To see this, recall that $\mathbf{r}_j = \mathbf{h} \mathbf{s}_j$ defines the scaled (invariant) position $\mathbf{s}_j$. It then follows from this and the definition of $\mathbf{k}$, equation (A 5), that

$$\begin{aligned} \mathbf{k} \cdot \mathbf{r}_j &= \sum_{\alpha} \left(\sum_{\mu} 2\pi h_{\mu\alpha}^{-1} K_{\mu} \right) \left(\sum_{\gamma} h_{\alpha\gamma} s_{j\gamma} \right) = \sum_{\alpha} \sum_{\mu} \sum_{\gamma} 2\pi h_{\mu\alpha}^{-1} K_{\mu} h_{\alpha\gamma} s_{j\gamma} \\ &= 2\pi \sum_{\mu} \sum_{\gamma} K_{\mu} s_{j\gamma} \sum_{\alpha} h_{\mu\alpha}^{-1} h_{\alpha\gamma} = 2\pi \mathbf{K} \cdot \mathbf{s}, \end{aligned} \quad (\text{A } 11)$$

the latter equality following from the fact that $\mathbf{h}^{-1} \mathbf{h} = \mathbf{1}$.

From equations (A 9), (A 3), (A 2) and (A 1) we arrive at the final result

$$P_{C_{\alpha\beta}}^F = \frac{1}{2V^2 \epsilon_0} \sum_{\mathbf{k} \neq \mathbf{0}} A_{\mathbf{k}} |Q_{\Sigma}|^2 \left\{ \delta_{\alpha\beta} - \left(\frac{1}{4\alpha^2} + \frac{1}{k^2} \right) 2k_{\alpha} k_{\beta} \right\} \quad (\text{A } 12)$$

as given in the main text (equation (27)).

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